PRE-REGISTRATION

Plant Architecture Inversion via Coherence Gradient Field Specification

Architectural Inversion of *Raphanus sativus* Produced by Pre-Specified Inversion of the Hydrotropic and Phototropic Coherence Gradient Field Before Germination

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Repository: github.com/Eric-Robert-Lawson/attractor-oncology
Pre-registration timestamp: 2026-03-13
Linked pre-registration DOI (LECA): 10.5281/zenodo.18986790
Document version: 1.0
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THE PRE-REGISTERED CLAIM

The organism is not the output of a genome alone. It is the output of a genome operating within a coherence gradient field. Change the field before germination and the genome executes a different developmental program. A different architectural form emerges. Same genome. Different field. Different organism.

This pre-registration specifies: the coherence gradient field conditions that will produce architectural inversion in *Raphanus sativus* (radish), the precise architectural outcome predicted, the scoring method, the control conditions required, and the success criteria — all locked before the experiment is run.

Primary prediction (P-INVERSION-1): A radish seed placed in an inverted coherence gradient field — mineral water gradient above, UV-A light source below, ambient light blocked — will germinate with root growing upward and shoot growing downward. Architecture Inversion Score (AIS) = 2 confirmed at T = 72 h and maintained at T = 168 h.

1. Background and Theoretical Basis

1.1. The Standard Model and Its Incompleteness

Contemporary biology operates on the model: **Genome → Organism.** The genome is treated as the blueprint. To change the organism, one changes the genome (CRISPR, directed evolution, synthetic biology).

This model is incomplete.

The complete model is: **Genome + Coherence Gradient Field → Organism.**

The genome contains *build capacity* — the capacity to build every structure the organism has built across its evolutionary history, and potentially more. The coherence gradient field is the *instruction set* that tells the genome which programs to express, where, and in which direction to navigate.

The organism that emerges is the resolution of the genome's build capacity within the field's instruction set.

1.2. Plant Architecture as Field Navigation

Standard plant architecture — root growing downward, shoot growing upward — is not genetically fixed. It is the attractor the genome resolves to within the standard coherence gradient field: water and minerals below, light above, gravity present.

Each axis navigates toward its coherence gradient:

- **Root:** Navigates toward water (hydrotropism, MIZ1/MIZ2 pathway), minerals, darkness, and physical medium. Negative phototropism: avoids light. Positive gravitropism: grows downward.
- **Shoot:** Navigates toward light (phototropism, phototropin pathway), CO_2, and warmth. Negative

gravitropism: grows upward.

Standard soil architecture is the attractor produced when all gradients point in the same direction: root gradients (water, minerals, darkness) are downward; shoot gradients (light) are upward. Gravity confirms both directions.

1.3. The Geometric Prediction

If plant architecture is a field-navigated attractor rather than a genetically fixed output, then:

Inverting the coherence gradient field before germination should produce an inverted architectural attractor.

Specifically: placing water above the seed and light below the seed — before germination begins — should cause the root to navigate upward (toward water) and the shoot to navigate downward (toward light). Gravity alone cannot override two simultaneous coherence gradient signals pointing in the opposite direction.

This prediction is geometrically derived from the attractor framework *before* the experiment is run. The conditions are specified from the geometry. The outcome is predicted from the conditions. The experiment confirms or falsifies the geometry.

1.4. Empirical Basis for Individual Mechanisms

Both component mechanisms are independently established in the published literature:

- **Hydrotropism overriding gravitropism:** Miao et al. (2025), *PNAS*. Root hydrotropism in *Arabidopsis* overrides gravitropism in the presence of a water gradient. The MIZ1 protein is the primary hydrotropism regulator. Upward root growth confirmed in moisture gradients pointing against gravity.
- **Root negative phototropism:** Galen et al. (2007), *Functional Ecology*. Roots exhibit negative phototropism to blue and UV-A wavelengths (phototropin receptors present in roots). Root grows away from light source.
- **Shoot positive phototropism:** Christie (2007), *Annual Review of Plant Biology*. Shoot phototropism well-established across all flowering plant species. Blue and UV-A light activates phototropin-1 and phototropin-2, driving shoot toward light.

This pre-registration predicts that **combining** these two independently established mechanisms — water above and light below — produces full architectural inversion (AIS = 2), not merely partial deflection of one axis. This combination and its outcome is the novel pre-specified prediction.

2. Pre-Registered Predictions

All predictions below are locked at the timestamp of this document's upload to Zenodo. No post-hoc modifications will be made to predictions. Amendments, if any, will be registered as separate versioned documents with new DOIs.

2.1. Primary Prediction: P-INVERSION-1

P-INVERSION-1 (Primary): In the EXP chamber (mineral water gradient above seed, UV-A LED below seed, ambient light blocked above), *Raphanus sativus* will germinate and grow with:

- Root directed **upward** (angle > 135ř from vertical downward) at T = 72 h.
- Shoot directed **downward** (angle < 45ř from vertical downward) at T = 72 h.
- Architecture Inversion Score **AIS = 2** at T = 72 h, maintained at T = 120 h and T = 168 h.

2.2. Control Predictions: P-INVERSION-2 through P-INVERSION-4

P-INVERSION-2 (C1 control): In the C1 chamber (standard conditions: agar only, ambient light above, no water gradient), AIS = 0 at all timepoints. Root grows downward. Shoot grows upward. Standard architecture confirmed.

P-INVERSION-3 (C4 hydrotropic control): In the C4 chamber (mineral water above seed, ambient light above, no LED below), root angle > 120ř at T = 72 h. Root grows upward driven by hydrotropism

alone. Shoot grows upward (ambient light above). Reproduces Miao et al. (2025) result in this setup. This control confirms the water gradient mechanism is functioning before interpreting EXP results.

P-INVERSION-4 (C3 phototropic control): In the C3 chamber (UV-A LED below, no water gradient, ambient light blocked), shoot grows downward (toward LED). Root grows downward (gravity dominant without water gradient counter-signal). Confirms the phototropic component alone is insufficient for full inversion. Both components required.

2.3. Architecture Inversion Score Definition

The Architecture Inversion Score (AIS) is the primary quantitative outcome measure. It is defined as follows:

AIS	Morphological Criterion	Interpretation
2	Root angle > 135ř AND shoot angle < 45ř (both axes fully inverted).	Full architectural inversion. P-INVERSION-1 confirmed.
1	One axis inverted (> 90ř from expected), other axis partially or not inverted.	Partial inversion. One gradient dominant, other insufficient.
0	Root angle < 45ř (downward) AND shoot angle > 135ř (upward).	Standard architecture. No inversion.
−1	Both axes inverted but in same direction (both up or both down).	Disorientation, not architectural inversion. Mechanism failure.

Angles are measured from the vertical (straight down = 0°, straight up = 180°) using a protractor held against the transparent chamber wall at each observation timepoint.

3. Experimental Design

3.1. Overview

Four chambers are run simultaneously in the same environment. All chambers use the same seed batch, same agar preparation, same temperature, and same room. The only variable between chambers is the configuration of the coherence gradient field.

3.2. The Four Chambers

ID	Water gradient	Light source	Purpose
EXP	Mineral water ABOVE seeds (30–40 mL)	UV-A LED below; ambient light blocked above	Experimental. Full inverted gradient field. Pre-registered prediction: AIS = 2.
C1	None (agar only, moist)	Ambient light above; no LED	Standard control. Normal architecture baseline.
C3	None (dry agar surface)	UV-A LED below; ambient light blocked above	Phototropic component only. No water gradient counter-signal.
C4	Mineral water ABOVE seeds (30–40 mL)	Ambient light above; no LED	Hydrotropic component only. Reproduces Miao et al. (2025).

3.3. Materials

Item	Cost (USD)	Source
Raphanus sativus seeds (Cherry Belle or French Breakfast)	$2–4	Garden store, Amazon
Agar powder (food-grade or bacteriological)	$3–8	Grocery store, Amazon
Mineral salts (KNO_3, KH_2PO_4, $MgSO_4$) OR bottled mineral water (Evian/Volvic)	$5–15	Hydroponics supplier, grocery
Transparent plastic cups or small jars (5×)	$3–8	Hardware store, dollar store
UV-A LED strip ∼365 nm	$10–20	Amazon
Black cardboard (light blocking)	$1–3	Art supply, dollar store
Distilled or filtered water	$1–2	Grocery store
Total	**$25–60**	

3.4. Mineral Solution Composition

Option A (from salts):

- 1 mM KNO_3
- 0.5 mM KH_2PO_4
- 0.2 mM $MgSO_4$
- pH 5.8–6.5, adjusted with dilute KOH if needed
- Prepared in distilled water

Option B (simplified): Commercially available mineral water (Evian, Volvic, or equivalent) used undiluted. Adequate ionic content for hydrotropism signalling.

3.5. Chamber Assembly

Each chamber consists of a transparent container (clear plastic cup, 300–500 mL, or equivalent glass jar) constructed as follows:

1. Pour 5–8 mL of 1% agar (1 g per 100 mL water, boiled and cooled to ∼50°C) into the base. Allow to set on a level surface for 15 minutes (confirmed solid by toothpick test).
2. Place 3–5 seeds on the agar surface, spaced ≥1.5 cm apart. Seeds sit ON the surface, not pressed in.
3. For EXP and C4: add 30–40 mL of mineral solution on top of the agar and seeds (the water gradient above the seeds).
4. For EXP and C3: install UV-A LED beneath the chamber, pointing upward. Cover the top of the chamber with black cardboard, confirmed light-tight.
5. For C4: top remains uncovered or covered with transparent film. Ambient room light from above.
6. For C1: standard orientation, ambient light above, no water gradient above seeds, no LED.

All chambers placed in the same room at 18–24°C. All chambers remain stationary throughout the experiment. Do not move chambers during observation.

3.6. Observation Protocol

Timepoint	Observation
T = 24 h	First germination check. Record: germination (%) per chamber. Photograph all chambers from side.
T = 48 h	Directional growth assessment. Record root and shoot direction (qualitative: up/down/sideways). Calculate preliminary AIS.
T = 72 h	**Primary outcome point.** Measure root and shoot angles with protractor. Calculate AIS for all chambers. Photograph all chambers with scale ruler.
T = 120 h	Secondary confirmation. AIS re-measured. Stability of inversion assessed.
T = 168 h	Final outcome. Remove one EXP seedling for close examination and side-by-side comparison with C1. Photograph on white background.

4. Success Criteria and Falsification

4.1. Confirmation of P-INVERSION-1

P-INVERSION-1 is confirmed if and only if:

1. EXP: AIS = 2 at T = 72 h **and** maintained at T = 168 h.
2. C1: AIS = 0 at all timepoints.
3. C4: Root angle > 120ř at T = 72 h. (The hydrotropism mechanism is confirmed functioning in this setup before EXP is interpreted.)

P-INVERSION-1 is **not** confirmed if: EXP achieves AIS = 2 but C4 does not show upward root (> 120ř). In this case the inversion result is real but the mechanism attributed (hydrotropism) is not confirmed by this setup. The result requires investigation before the mechanism claim is made.

4.2. Falsification Criteria

The geometric prediction is falsified if:

- EXP: AIS = 0 at T = 72 h (standard architecture despite inverted field). This would indicate gravity dominates over both combined coherence gradients at this field strength.
- EXP: AIS = 1 at T = 72 h (partial inversion only). This is a partial result requiring further investigation of field strength. Not a confirmation of the full geometric prediction.
- EXP: AIS = −1 (both axes disoriented). This would indicate the combined field produced developmental confusion rather than inversion. A distinct and informative failure mode.

5. Reproducibility Requirements

For any replication of this experiment to be considered valid and comparable to this pre-registration, the following minimum documentation must be provided:

1. Species and seed source (cultivar, supplier).
2. Seed viability test result (germination rate from pre-experiment test).
3. Agar concentration (g per 100 mL water).
4. Water gradient composition (salt concentrations or mineral water brand).
5. LED specification (wavelength range in nm, power in mW or W, distance from seed position in cm).
6. Light blocking method and confirmation (material used, how light leak was checked).
7. Temperature during experiment.
8. All four chambers run simultaneously in the same environment.

9. AIS scores at T = 72 h, T = 120 h, T = 168 h for all chambers.
10. At least one photograph per chamber per timepoint, labelled with chamber ID, timepoint, and date.

6. Relationship to the Broader Framework

This experiment is one component of a larger pre-registered research program (OrganismCore, DOI: 10.5281/zenodo.1898679 testing the geometric claim that organisms are the output of genomes operating within coherence gradient fields.

The plant inversion experiment is the first *prospective* test of this claim — a pre-specified prediction confirmed by experiment rather than a retrospective explanation of already-observed phenomena.

The retrospective confirmation record (14 cases across all kingdoms, documented in the OrganismCore repository) is available at: github.com/Eric-Robert-Lawson/attractor-oncology

7. Declaration

This document is submitted as a pre-registration. The experiment described has **not** been run at the time of this document's creation and upload. All predictions are made before any experimental data exists. Results, when obtained, will be reported in a separate document linked to this pre-registration by DOI. No modifications will be made to this document's predictions after upload. Amendments will be versioned and separately registered.

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The field came first. The genome was always listening. The architecture was never fixed. The inversion was always possible.
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